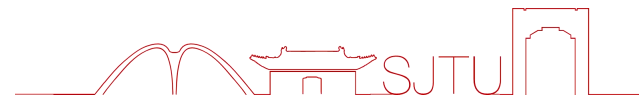




上海交通大学
SHANGHAI JIAO TONG UNIVERSITY



Attention is all you need

Transformer

2023年12月21日

饮水思源 · 爱国荣校



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神经网络

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Transformer

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Python实现

01

神经网络

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神经网络(Neural Networks)是一系列模拟人脑工作机制的机器学习(Machine Learning)技术。

挖掘出隐藏在一堆数据背后的规律的能力对数据科学家至关重要, 而神经网络技术很适合提取图片、视频和音频里面的特征。





输入

运算

输出

x

$F(x)$

y

向量, 时序序列,
图片, 视频流....

CNN, RNN,
Transformer....

....



How it works



以MPL为例

前馈 forward : **prediction**

$$Y = \text{HidL}_4(\text{HidL}_3(\text{HidL}_2(\text{HidL}_1(x))))$$

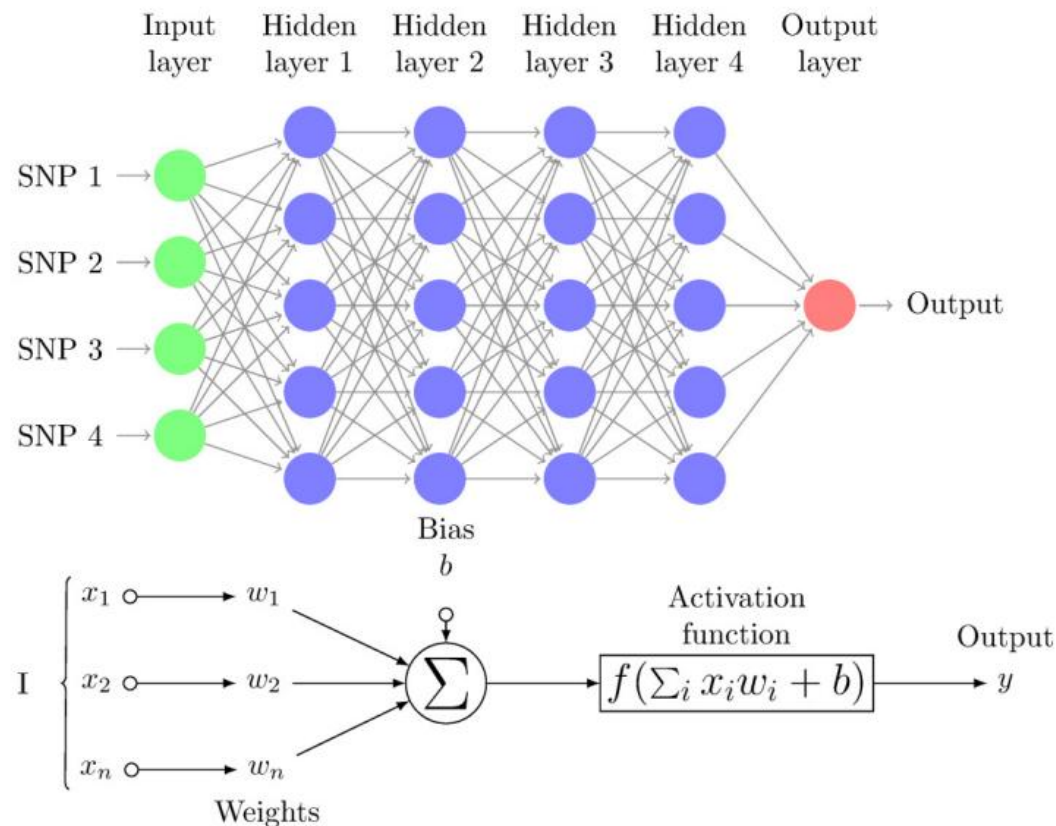
$$\text{HidL}_i(x)$$

$$= \text{concat}(\text{RELU}(\Sigma(w_j x_j + b_j)))$$

反馈 backward : **learning**

$$Y_{\text{real}} - Y_{\text{pred}}$$

$$\left. \frac{\partial y}{\partial w_{1j}} \right|_{x=x_j} = H'_4(x)H'_3(x)H'_2(x)H'_1(x)x$$



02

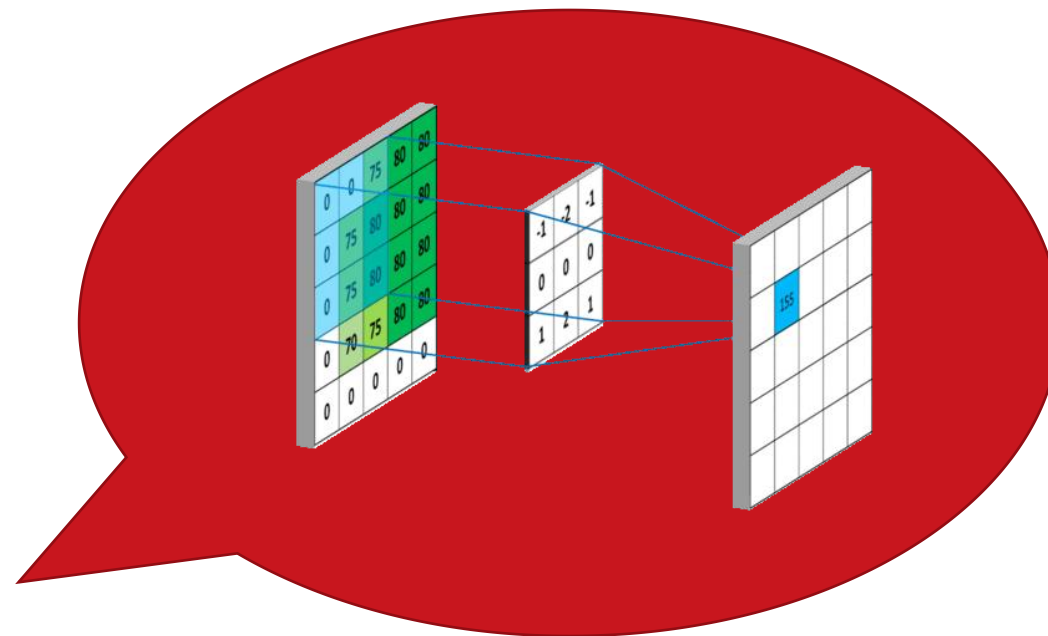
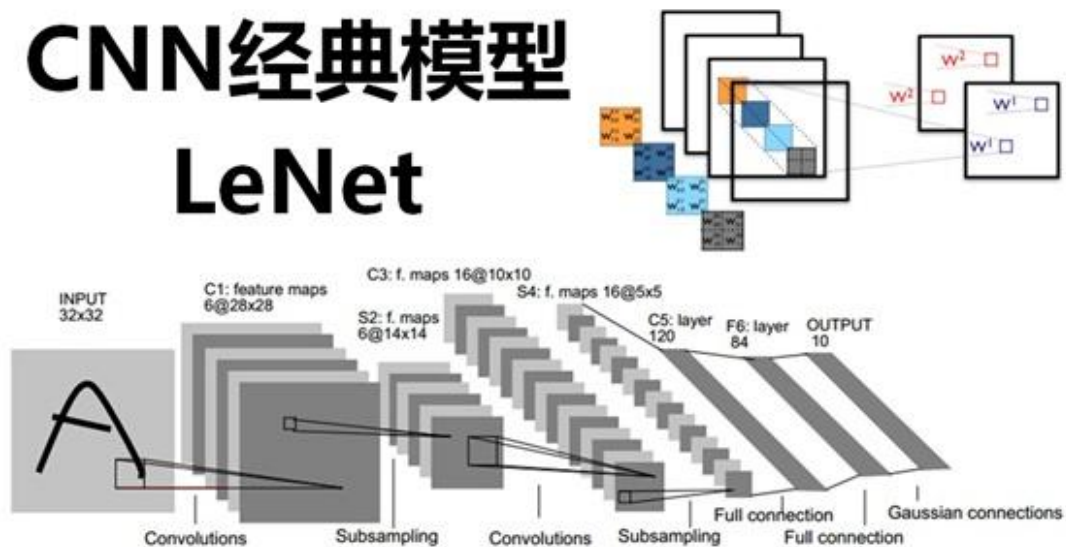
Transformer

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■ CNN (convolutional neural network)

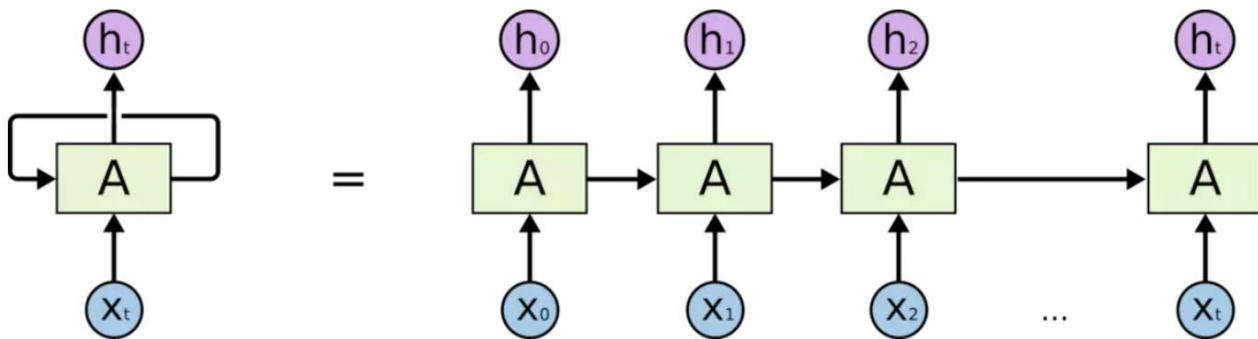
CNN经典模型 LeNet



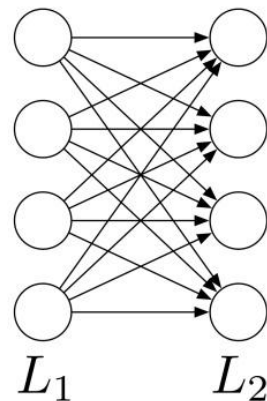
2010-2017计算机视觉CV等领域

RNN (Recurrent Neural Networks)

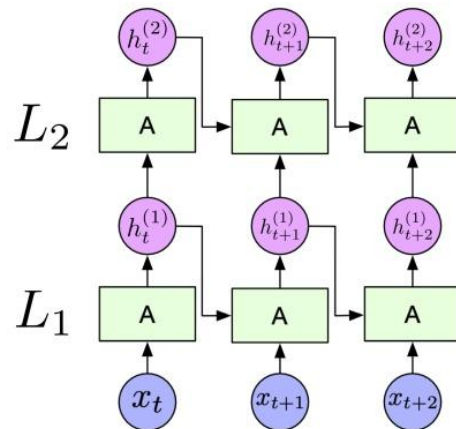
- RNN 是包含循环的网络，可以把信息从上一步传递到下一步。



RNN的循环展开之后其实就是同一个网络复制多份，次序连接进行信息传递。



FCN



RNN 乎 @邵智鹏

2014-2017自然语言处理NLP等领域





Transformer

Attention is all you need





Transformer

Attention is all you need

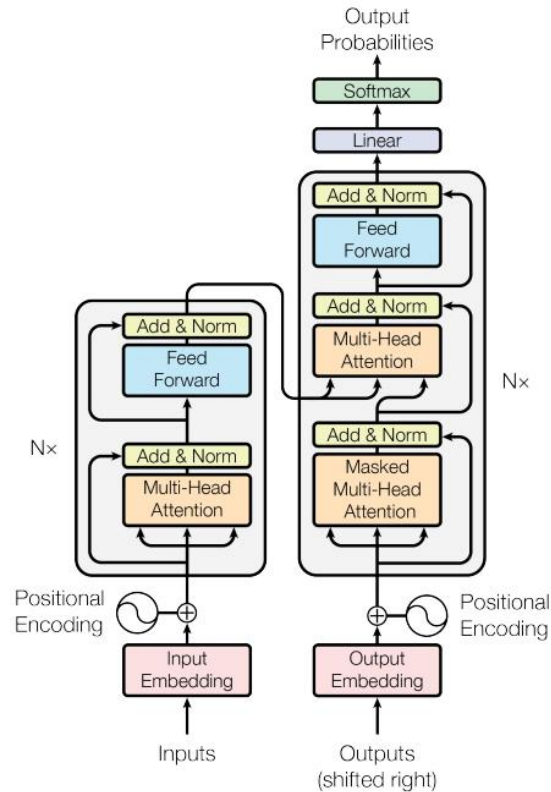


Figure 1: The Transformer - model architecture.

Attention Is All You Need

Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google Research
usz@google.com

Llion Jones*
Google Research
llion@google.com

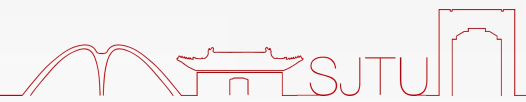
Aidan N. Gomez* †
University of Toronto
aidan@cs.toronto.edu

Łukasz Kaiser*
Google Brain
lukaszkaizer@google.com

Illia Polosukhin* ‡
illia.polosukhin@gmail.com

Abstract

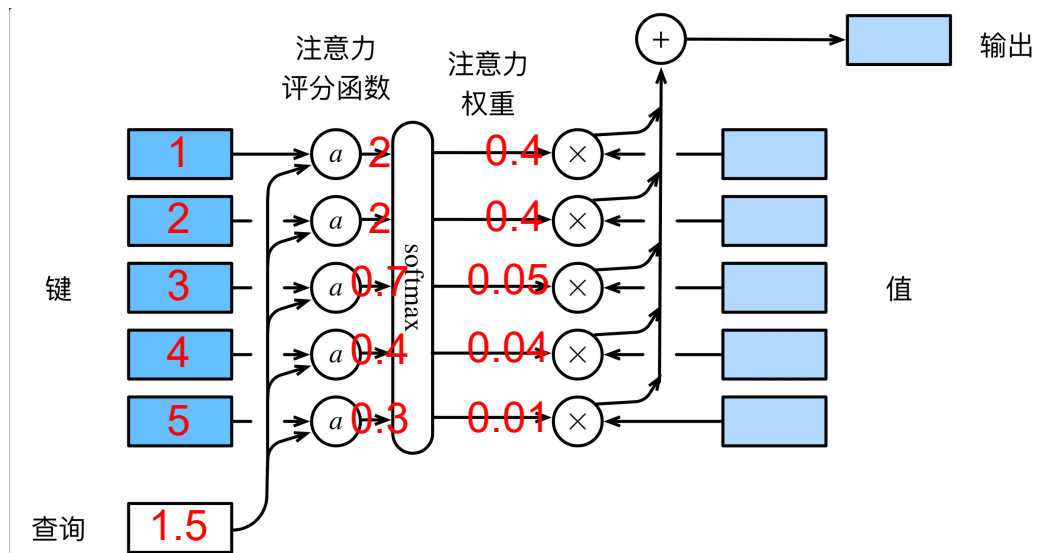
The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.





Attention

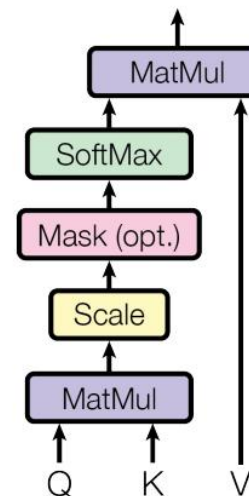
What is attention



$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

d_k 是 Q, K 矩阵的列数，即向量维度

Scaled Dot-Product Attention





Tricks

1. 多头;
2. 残差连接;
3. positional encoding

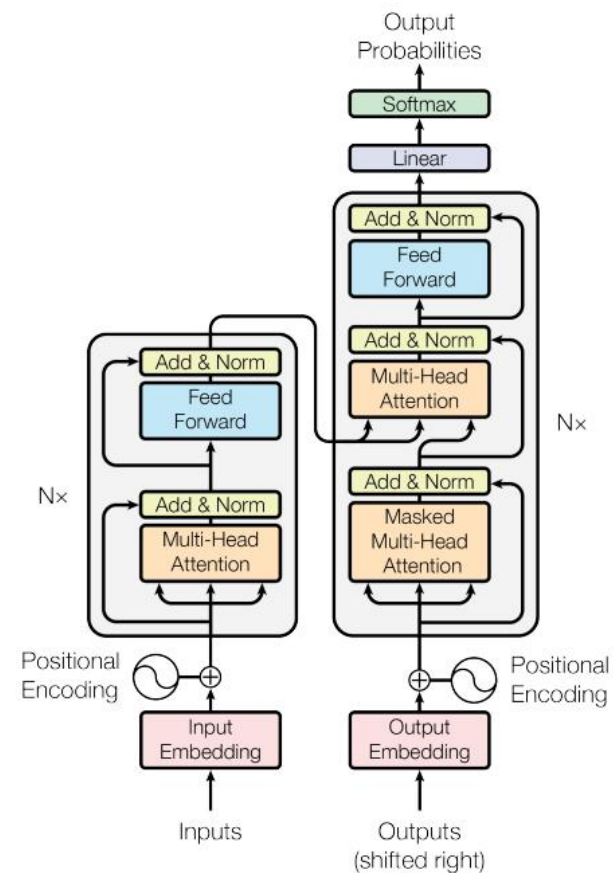


Figure 1: The Transformer - model architecture.





Transformer

Multi-Head Attention 多头

将初始向量投影到不同的空间上
从多方面提取特征

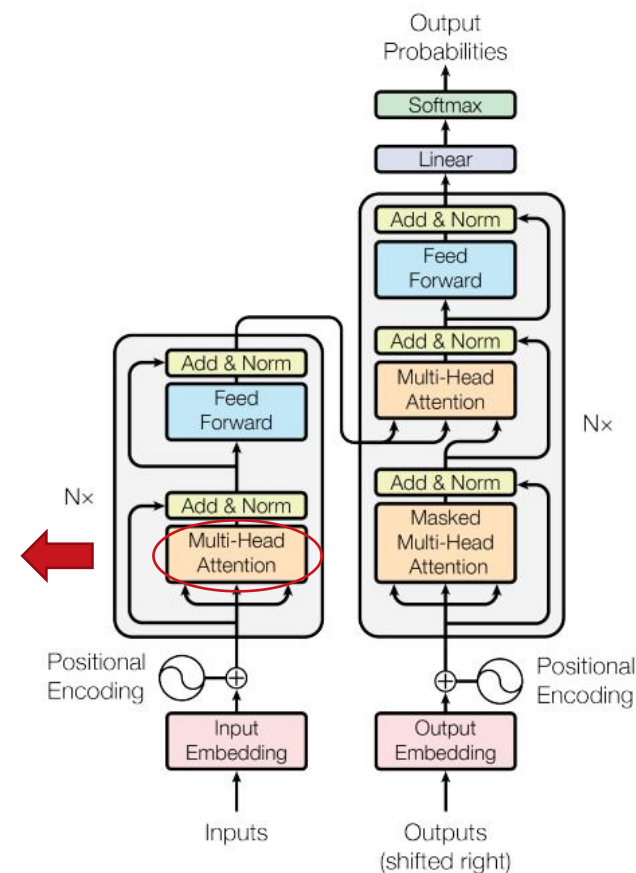
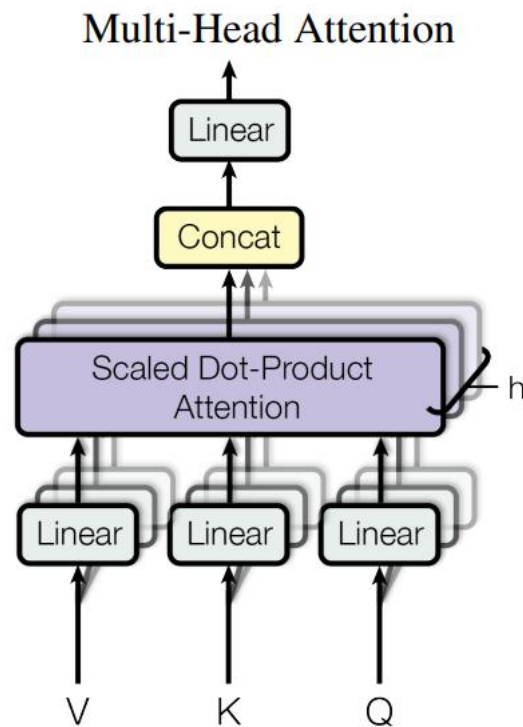


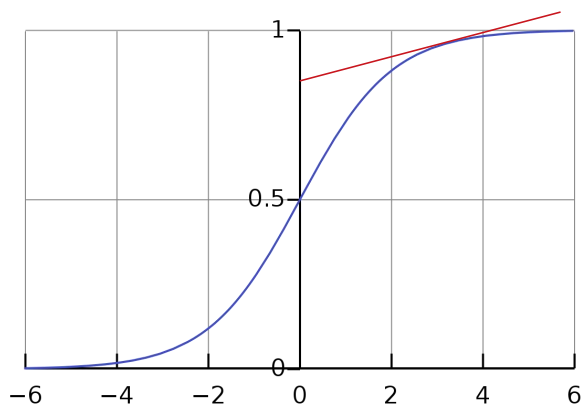
Figure 1: The Transformer - model architecture.



残差连接

$$\left. \frac{\partial y}{\partial w_{1j}} \right|_{x=x_j} = H'_4(x)H'_3(x)H'_2(x)H'_1(x)x \rightarrow 0.1^4$$

$$H_i(x) \rightarrow H_i(x) + x \rightarrow (0.1 + 1)^4$$



防止梯度消失

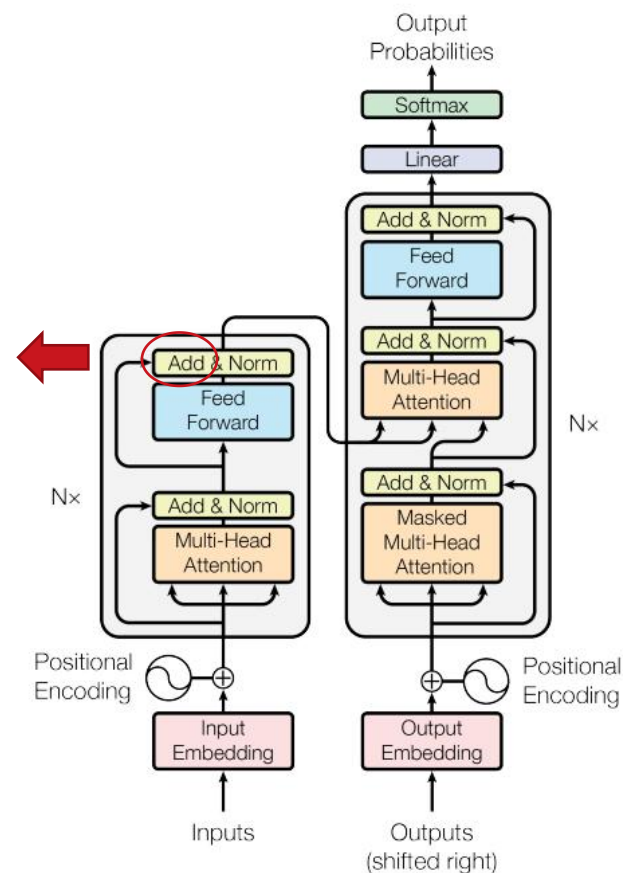


Figure 1: The Transformer - model architecture.

”

positional encoding

Transformer 不采用 RNN 的结构，而是使用全局信息，不能利用单词的顺序信息，而这部分信息对于 NLP 来说非常重要。

$$PE_{(pos,2i)} = \sin(pos/10000^{2i/d})$$
$$PE_{(pos,2i+1)} = \cos(pos/10000^{2i/d})$$

加入序列顺序信息

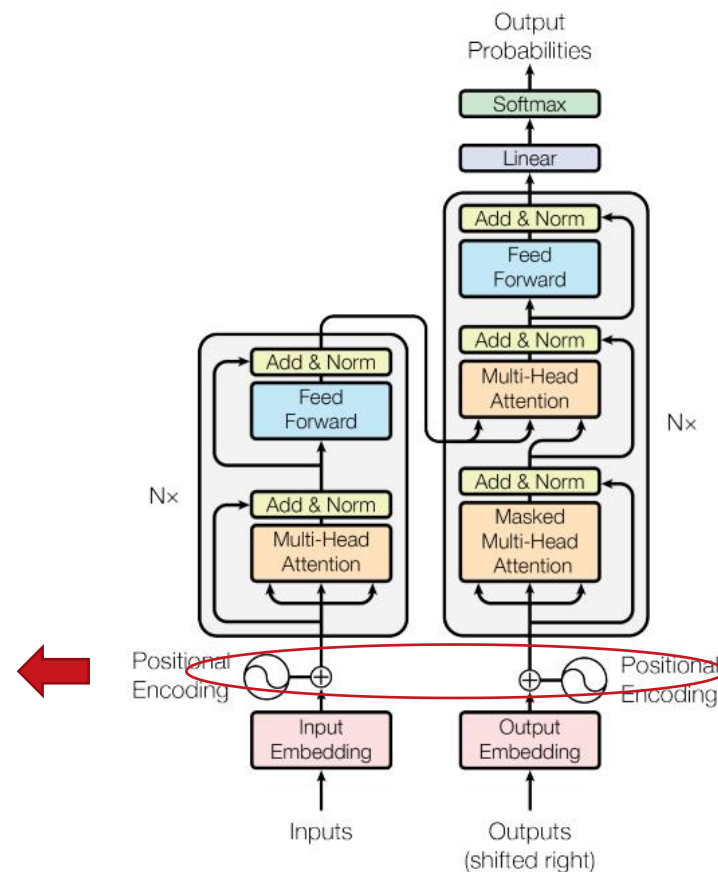


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Transformer

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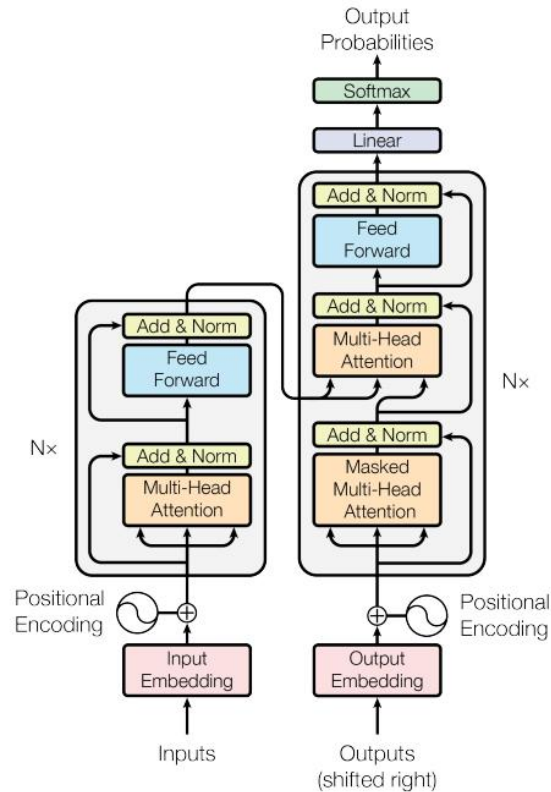


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Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google Research
usz@google.com

Llion Jones*
Google Research
llion@google.com

Aidan N. Gomez* †
University of Toronto
aidan@cs.toronto.edu

Łukasz Kaiser*
Google Brain
lukaszkaizer@google.com

Illia Polosukhin* ‡
illia.polosukhin@gmail.com

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.



03

Python实现

请在此输入文字说明





Anaconda+虚拟环境

安装库(torch和其它)

运行截图

代码对应结构图, train.py文件开头参数的意义

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感谢聆听

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